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COMPUTER NETWORKS

Ex.No.4	**a) Setup and configure a LAN (Local area network) using a Switch and Ethernet cables in your lab.** 1. Connect 3-4 host machines to a switch. 2. Assign ip addresses to each host machine. 3. Check the connectivity between the machines by using ping command. 4. Share and access files and folder across the machines of the LAN.
Date:	

LAN:

A local area network (LAN) is a collection of devices connected together in one physical location, such as a building, office, or home. A LAN can be small or large, ranging from a home network with one user to an enterprise network with thousands of users and devices in an office or school.

Regardless of size, a LAN's single defining characteristic is that it connects devices that are in a single, limited area.

Set Up a LAN (Local Area Network) Directly

How to Set Up a LAN (Local Area Network)

Step 1: Plan the Network
Plan and design an appropriate network topology based on the network requirements and the physical location of the devices.

Step 2: Gather the Required Devices
Prepare the following hardware components:

- 4 Computers (PCs)
- 1 Network Switch (8-port, 16-port, or 24-port)
- 4 Ethernet (UTP) Cables

Step 3: Connect the Devices
Connect each computer to the network switch using Ethernet cables. Plug one end of the Ethernet cable into the Ethernet port of the computer and the other end into an available port on the switch. Ensure that the link LEDs on the switch and the computers are ON, indicating successful connections.

Step 4: Assign IP Addresses to the PCs

1. Log in to each computer using an **Administrator** or **Owner** account.
2. Open **Control Panel → Network and Internet → Network and Sharing Center**.
3. Click **Change Adapter Settings**.
4. Right-click **Local Area Connection** (or **Ethernet**) and select **Properties**.
5. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.
6. Select the **Use the following IP address** option.
7. Enter the IP address and subnet mask for each computer as shown below:

Computer	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0

Click **OK** to save the configuration. Repeat the same procedure for all the PCs connected to the switch.

Step 5: Configure the Network Switch

1. Connect a computer to the switch using an Ethernet cable.
2. Open a web browser and enter the IP address of the switch in the address bar to access the switch's management interface.
3. Log in using the appropriate **username** and **password**.
4. Configure the basic settings of the switch.
5. Assign the following management IP address to the switch:
6.

Device	IP Address	Subnet Mask
Switch	10.1.1.5	255.0.0.0

Save the configuration.

Step 6: Verify Network Connectivity

Open the **Command Prompt** on each computer and use the **ping** command to test connectivity with the other computers and the switch.

Example:

- `ping 10.1.1.2`
- `ping 10.1.1.3`
- `ping 10.1.1.4`
- `ping 10.1.1.5`

If replies are received without packet loss, the network has been configured successfully.

Step 7: Share a Folder

1. Create or select a folder to be shared.
2. Right-click the folder and select **Properties**.
3. Open the **Sharing** tab.
4. Click **Share**.
5. Select **Everyone** from the list of users.
6. Click **Share** and then **Done**.

The selected folder is now accessible to all computers on the same LAN.

Step 8: Access the Shared Folder

1. Open **File Explorer** on another computer connected to the LAN.
2. In the address bar, type the IP address or computer name of the shared computer. For example:
    - `\\10.1.1.1`
    - `\\PC1`
3. Press **Enter** to view the shared folder.
4. Open the folder and verify that files can be accessed successfully.

Procedure to Set Up a LAN Using Cisco Packet Tracer

Step 1: Design the Network Topology

1. Open **Cisco Packet Tracer**.
2. From the **Network Devices** section, select **Switches** and drag one **2960 Switch** onto the workspace.
3. From the **End Devices** section, drag four **PCs** onto the workspace.
4. Rename the devices as **PC0, PC1, PC2, and PC3**.

Step 2: Connect the Devices

1. Click **Connections** (Lightning Bolt icon).
2. Select **Copper Straight-Through Cable**.
3. Connect the devices as follows:

Source Device	Interface	Destination Device	Interface
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1	FastEthernet0	Switch0	FastEthernet0/2
PC2	FastEthernet0	Switch0	FastEthernet0/3
PC3	FastEthernet0	Switch0	FastEthernet0/4

4. Wait until the link LEDs turn **green**, indicating that the devices are successfully connected.

Step 3: Assign IP Addresses to the PCs
For each PC:

1. Click the PC.
2. Select the **Desktop** tab.
3. Click **IP Configuration**.
4. Enter the following IP addresses:

Device	IP Address	Subnet Mask
PC0	10.1.1.1	255.0.0.0
PC1	10.1.1.2	255.0.0.0
PC2	10.1.1.3	255.0.0.0
PC3	10.1.1.4	255.0.0.0

Leave the **Default Gateway** blank because no router is used in this LAN.

Step 4: Verify IP Configuration
On each PC:

1. Open **Desktop**.
2. Select **Command Prompt**.
3. Execute the following command:

```
ipconfig
```
Verify that the configured IP address and subnet mask are displayed correctly.

Step 5: Test Network Connectivity
From **PC0**, open **Command Prompt** and execute:

```
ping 10.1.1.2
```
Similarly, test connectivity with the remaining PCs:

```
ping 10.1.1.3
```

```
ping 10.1.1.4
```
Repeat the ping test from each PC to ensure that all hosts can communicate successfully.

Step 6: Verify Packet Transmission Using Simulation Mode

1. Click **Simulation** mode at the bottom-right corner of Packet Tracer.
2. Select the **Add Simple PDU** tool (Envelope icon).
3. Click **PC0** as the source device.
4. Click **PC3** as the destination device.
5. Click **Auto Capture/Play**.

Observe the ICMP Echo Request and Echo Reply packets travelling through the switch.

Step 7: Configure an FTP Server for File Sharing
Since Packet Tracer does not support Windows folder sharing, use an FTP server to simulate file sharing.

1. Drag one **Server** from **End Devices** into the workspace.
2. Connect the server to the switch using a **Copper Straight-Through Cable**.
3. Configure the server with:

Device	IP Address	Subnet Mask
Server0	10.1.1.5	255.0.0.0

4. Open **Server0**.
5. Select **Services → FTP**.
6. Turn the FTP service **ON**.
7. Create a username and password (for example, **student / student123**).

Step 8: Access the FTP Server
On **PC0**:

1. Open **Desktop → Command Prompt**.
2. Connect to the FTP server using:

```
ftp 10.1.1.5
```

3. Enter the configured username and password.
4. Execute FTP commands such as:

```
dir
get sample.txt
put report.txt
```

Verify that files can be uploaded to and downloaded from the FTP server.

Step 9: Save the Project

1. Click **File → Save As**.
2. Save the Packet Tracer file with an appropriate name, for example:

```
LAN_Configuration.pkt
```

The LAN configuration is now complete.

RESULT: The LAN has been configured using switch and an ethernet. The study of the LAN and switch was also studied.

